Robust Data-Driven Method for Data and Label Noise Mitigation in Synchronous Machine Winding Short-Circuit Fault Diagnosis

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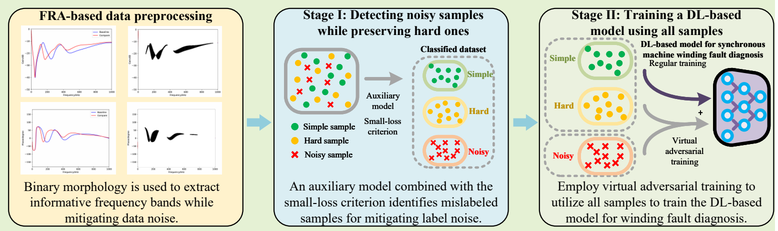
***Abstract*— Winding short-circuit (SC) faults pose a critical challenge in synchronous machines, where accurate and timely diagnosis is essential to ensure operational reliability and prevent cascading failures in the power system. Conventional inspection of synchronous machine windings relies heavily on labor-intensive expert assessment, which is often subjective and inefficient. Although recent data-driven approaches have advanced SC fault diagnosis, they require training samples with low noise and accurate labels, incurring prohibitive costs. To address these limitations, this study proposes a robust data-driven method for joint mitigation of data and label noise in synchronous machine winding SC fault diagnosis. To mitigate data noise, binary morphology is used to preprocess frequency response analysis (FRA) data before input into the data-driven model, extracting informative frequency bands while suppressing perturbations. For label noise, an auxiliary model combined with the small-loss criterion identifies mislabeled samples in the training dataset, followed by virtual adversarial training (VAT) to utilize all samples, treating mislabeled samples as unlabeled, in a semi-supervised learning mode. The proposed method is validated on 5 kVA and 7.5 kVA synchronous machine FRA datasets. Under the controlled combined noise settings, it maintains accuracies above 90% even when the training data contain up to 40% incorrect labels and additive Gaussian data noise down to 10 dB SNR, with the remaining misclassifications mainly occurring among inherently similar fault categories.**



***Index Terms*— Data noise, fault diagnosis, frequency response analysis, label noise, synchronous machine, winding fault.**

I. Introduction

SYNCHRONOUS machines are indispensable in modern power systems, forming the backbone of large-scale energy conversion [1]. However, winding faults remain the dominant failure mode, accounting for nearly two-thirds of synchronous machine breakdowns, with short-circuit (SC) faults such as inter-turn (ITSC) and ground short-circuits (GSC) being the most common [1], [2]. These faults, often induced by insulation degradation, transient overvoltages, and electromechanical stresses, can escalate rapidly, leading to overheating, torque oscillations, and severe instability in the power system if not detected in time [3], [4]. Therefore, reliable diagnosis of winding SC faults is essential to ensure operational safety and availability.

Existing diagnostic strategies can be broadly categorized into online monitoring and offline testing [4]–[7]. Online methods, such as vibration or thermal monitoring [8]–[10], require high-precision sensors that are costly and difficult to retrofit into operating machines. In contrast, offline methods, including inductive impedance [11], repetitive surge oscilloscopes (RSO) [12], and frequency response analysis (FRA) [1], are generally more cost-effective, as they avoid additional sensor installation, and often achieve higher diagnostic accuracy by being less constrained by strict online measurement conditions.

FRA is a well-established technique in transformer fault diagnosis, demonstrating particular efficacy for detecting winding deformations [13]–[15]. Motivated by the structural similarities between transformer and synchronous machine windings, recent studies have extended FRA to the latter [1], [3], [16]–[20]. However, despite its high sensitivity to winding faults, inherent limitations have prevented FRA from

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becoming a mainstream diagnostic method for synchronous machines [20]. Two critical challenges persist: (1) susceptibility to measurement perturbations (i.e., data noise), such as cabling-induced noise, measurement error of sensors, and electromagnetic interference in the environment; and (2) reliance on subjective expert interpretation of FRA data [13].

In response to these limitations, researchers have enhanced FRA by incorporating data-driven methods to suppress data noise and introduce intelligent fault diagnosis. For the former, researchers leverage the inherent generalization and robustness of diverse data-driven methods to mitigate input data noise. For the latter, these methods replace subjective expert interpretation of FRA data with objective and automated judgments. Specifically, Refs. [21]–[23] integrate FRA-based mathematical-statistical indicators with various support vector machines (SVMs) or other machine learning (ML) methods to classify fault type, severity, and location; Refs. [24], [25] introduce deep learning (DL) methods, including explainable AI (XAI) to improve interpretability and lifelong learning to enable continuous model adaptation. However, most existing methods are evaluated only on artificially noisy test sets to demonstrate empirical noise robustness, without systematic analysis of the underlying noise-suppression mechanism.

In practical engineering applications, a pressing challenge arises from the prevalence of label noise in real-world datasets [26], [27]. This issue is critical, as the effectiveness of data-driven methods depends on accurate labels, which require annotations by domain-specific experts [26]. Although expert-provided “strong labels” are highly reliable, their acquisition is often cost-prohibitive in practice [28]. As a viable alternative, “weak labels” obtained from non-expert engineers typically present a more accessible and cost-effective solution. However, such weakly labeled datasets inevitably contain incorrect labels, commonly termed label noise, primarily due to three factors: (1) Coupling between data and label noise, meaning that even experts may mislabel samples affected by data noise; (2) Similarity among different winding faults, which induces ambiguous decision boundaries; (3) Human factors, whereby annotators may give incorrect labels due to distraction or fatigue. Data-driven methods, capable of approximating arbitrarily complex functions, are particularly vulnerable to label noise. Without proper label noise mitigation, models can learn incorrect input-output mappings, leading to erroneous assessments. This issue not only compromises the model’s ability to generalize and its predictive accuracy but may, in severe instances, threaten the safety of the power system [26].

Recent advances have further enriched the study of label noise mitigation in fault diagnosis. Ref. [29] introduces the extended invariant risk minimization (EIRM) framework, which integrates flat-minima optimization with invariant feature learning to jointly address label noise and domain shift. Ref. [30] proposes a meta-self-training strategy based on a teacher-student network, leveraging meta-learning to reduce the confirmation bias of self-training and improve label correction under noisy conditions. Ref. [31] introduces an improved robust auxiliary classifier for bearing fault diagnosis in wind turbine gearboxes, which effectively generates realistic fault data and improves classification accuracy under limited and noisy label conditions. Collectively, these studies underscore the growing importance of noise-robust learning frameworks in practical fault diagnosis.

Based on a comprehensive review of existing references, it can be concluded that current data-driven methods integrating FRA suffer from two primary problems:

1) Existing approaches primarily rely on diverse classifiers and the presumed robustness of ML/DL-based models to handle input data noise. However, their effectiveness is highly architecture-dependent, and they lack systematic theoretical analysis to explain the mechanisms of data-noise suppression, resulting in limited adaptability and interpretability.
2) Most prior studies assume the availability of perfectly accurate labels, an assumption rarely satisfied in practical scenarios. In reality, weak labels are prevalent, and the presence of mislabeled samples significantly undermines the reliability and applicability of existing data-driven methods for winding fault diagnosis.

To address these challenges, this study proposes a robust data-driven method for synchronous machine winding SC fault diagnosis that explicitly mitigates both data and label noise. The main contributions are as follows:

1) We present a framework that integrates binary morphology with two-stage weakly supervised learning (WSL) to investigate the coupling between data and label noise in FRA-based fault diagnosis. In contrast to prior studies, our work provides experimental analysis and supporting discussion to examine the joint effects of data and label noise, thereby offering an enhanced perspective on noise interaction.
2) We design an integrated pipeline that combines binary morphology for preprocessing FRA data with the small-loss criterion and virtual adversarial training (VAT) in a semi-supervised learning mode. This design not only suppresses data noise and identifies mislabeled samples but also exploits mislabeled samples to reduce reliance on strong labels, thereby enhancing diagnostic reliability.
3) We demonstrate the complementary synergy between binary morphology and VAT. Binary morphology preprocesses FRA data to suppress data noise but may leave residual speckles, whereas VAT enhances robustness by smoothing the decision boundaries against such speckles, thereby ensuring that minor perturbations do not compromise the accuracy of winding SC fault diagnosis.

The remainder of this study is organized as follows: Section II introduces the principles of FRA, binary morphology, and WSL. Section III describes the experimental dataset. Section IV presents the results and ablation studies. Section V discusses limitations and industrial applicability. Section VI concludes this study.

II. METHODOLOGY

This section first presents the use of FRA for detecting winding SC faults, then describes the application of binary morphology to preprocess FRA data as input for the subsequent model, and finally introduces the proposed two-stage WSL framework.

A. Basic Principle of FRA

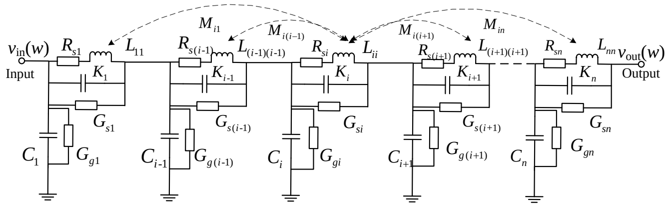

Fig. 1: Equivalent circuit model of a synchronous machine winding [32].

At high frequencies (above 1 kHz), synchronous machine windings can be modeled as an equivalent circuit composed of cascading RLC units [32], as shown in Fig. 1. The occurrence of winding faults induces relative changes in the internal distributed parameters, particularly the inductance and capacitance values. These alterations consequently modify the system's transfer function, which is manifested as changes in the FRA signature, commonly referred to as FRA data, including FRA gain and phase data. The winding's FRA gain data $H(\omega)$ and FRA phase data $\phi(\omega)$ can be acquired by applying excitation signals $\vec{v}_{\text{in}}(\omega)$ at one terminal and measuring the corresponding response signals $\vec{v}_{\text{out}}(\omega)$ at the other terminal. Then, by comparing the measured FRA data with normal (i.e., baseline) FRA data and analyzing deviations in both the resonant frequency and magnitude, winding conditions can be assessed [14], as shown in Fig. 2.

$$H(\omega) = 20\log_{10}\left|\frac{\vec{v}_{\text{out}}(\omega)}{\vec{v}_{\text{in}}(\omega)}\right| \text{dB} \tag{1}$$

$$\phi(\omega) = \frac{180}{\pi}\arg\left|\frac{\vec{v}_{\text{out}}(\omega)}{\vec{v}_{\text{in}}(\omega)}\right| \tag{2}$$

where $\vec{v}_{\text{in}}(\omega)$ represents the excitation signal (voltage or current), $\vec{v}_{\text{out}}(\omega)$ represents the corresponding response signal (voltage or current), $H(\omega)$ represents the FRA gain data, $\phi(\omega)$ represents FRA phase data, and ω is frequency. Referring to Refs. [1], [3], [4], [16], [24], [25], this study uses a frequency range from 1 kHz to 1 MHz.

B. Integration of Binary Morphology with FRA

A primary challenge in FRA application arises from data noise, which may perturb high-sensitivity frequency bands and confuse fault diagnosis [13].

To address this issue, binary morphology, a nonlinear image processing technique, is employed to extract frequency bands exhibiting informative FRA gain and phase variations. Specifically, the process begins by converting the difference between the two FRA data (i.e., the measured and normal FRA data) into two binary images (i.e., gain and phase images), represented by two 0-1 matrices. In these matrices, the rows correspond to discrete gain or phase levels and the columns to frequency points. Pixels between the measured and normal FRA data are assigned a value of 1 (foreground), while all others are set to 0 (background). Subsequently, morphological erosion is applied to eliminate noise and minor burrs, followed

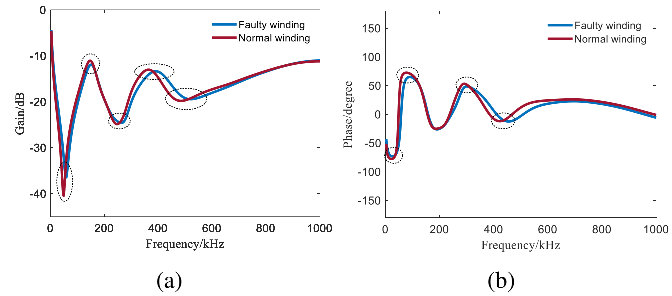

Fig. 2: FRA data of a synchronous machine winding before and after a winding SC fault. (a) FRA gain data. (b) FRA phase data.

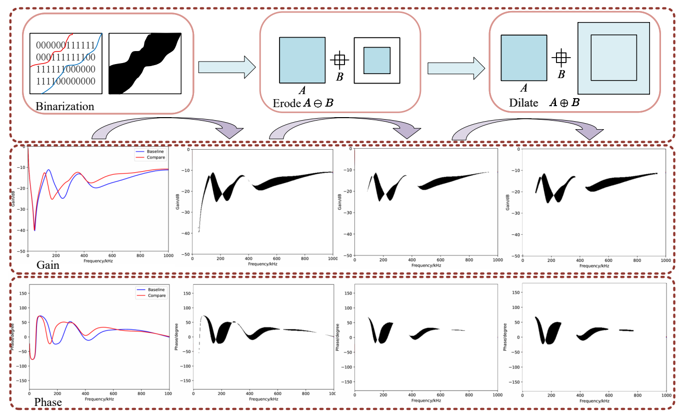

Fig. 3: Process of binarization, erosion, and dilation operation.

by a dilation operation to restore legitimate features that may have been excessively removed during the erosion process. The complete process is illustrated in Fig. 3, where the FRA gain and phase data, after being processed by binary morphology, serve as input to the subsequent model.

$$(A \oplus B)(x,y) = \max_{(s,t)\in B}\left[A(x-s,y-t)\right] \tag{3}$$

$$(A \ominus B)(x,y) = \min_{(s,t)\in B}\left[A(x+s,y+t)\right] \tag{4}$$

where A is the input binary image, B is a compact structuring element (i.e., a compact binary kernel), and $(s,t) \in B$ enumerates the relative offsets of the structuring element relative to its origin. The dilation operation $\oplus$ computes the maximum value over the translated neighborhood of A induced by B at position (x,y), whereas erosion $A \ominus B$ takes the minimum value.

C. Implementation of the Two-Stage WSL Framework

Existing data-driven methods for winding SC fault diagnosis depend on training samples with strong labels [4], [24], [25]. However, manual annotation by non-experts inevitably introduces label noise. To address this, WSL methods, particularly those utilizing the small-loss criterion, have been widely adopted [26]. This criterion leverages the observation that deep neural networks exhibit a memorization effect, prioritizing the learning of clean samples (i.e., samples with correct labels) over noisy ones (i.e., samples with wrong labels) during early training stages [27]. Consequently, samples maintaining consistently low loss are retained as clean, while high-loss samples are discarded (see Fig. 4). Clean samples can be further categorized into simple and hard samples. Simple samples yield low losses because the model can easily capture their fault characteristics. Conversely, hard samples with correct labels exhibit higher losses due to feature ambiguity or overlapping boundaries. These hard samples are critical for generalization, as they enable the model to discern subtle inter-class distinctions [26]. While the small-loss criterion mitigates the impact of mislabeled samples, it suffers from a critical limitation: it frequently misidentifies informative hard samples as noisy ones, as illustrated in Fig. 4. This exclusion, combined with the complete discarding of noisy samples, reduces data utilization and compromises the model's robustness and generalization capability.

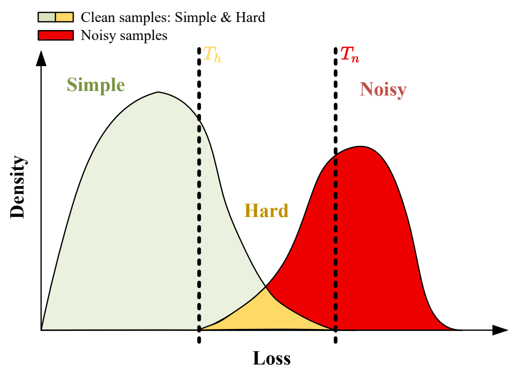

Fig. 4: Loss distribution of clean and noisy samples.

To address this issue, this study introduces a two-stage WSL framework that facilitates stable training in the presence of label noise. In Stage I, an auxiliary ML model leverages the full dynamic training loss to discriminate between noisy and hard samples, thereby preserving informative samples while mitigating the adverse effects of label noise. In Stage II, VAT is applied to all samples to train a DL-based model with enhanced generalization and robustness. The overall framework is illustrated in Fig. 5.

Stage I: Detecting noisy samples while preserving hard ones

First, the DL-based winding fault model $\mathcal{M}_0$ is pretrained on the noisy dataset $\mathcal{D}$, while recording per-sample dynamic training loss L_0. Using the final epoch losses $L_{\text{final epoch}}$ and the small-loss criterion, samples are split into a simple dataset $\mathcal{D}_s$ and a hard/noisy dataset $\mathcal{D}_{h/n}$. Second, an artificial noisy dataset $\mathcal{D}_a$ is generated from $\mathcal{D}_s$ by randomly altering the labels of a proportion η_a of its samples. Third, a new model $\mathcal{M}_1$ is retrained on the artificial noisy dataset $\mathcal{D}_a$, while recording per-sample dynamic training losses L_a. Subsequently, the small-loss criterion is applied to identify simple samples $\eta_{a,s}$, and samples with altered labels are designated as noisy samples $\eta_{a,n}$ while the remaining samples are hard samples $\eta_{a,h}$. Fourth, an auxiliary model $\mathcal{M}_t$ (this study uses SVM) is trained on $L_{a,h/n}$ with labels $\eta_{a,h/n}$ to distinguish hard and noisy samples. Finally, the trained auxiliary model

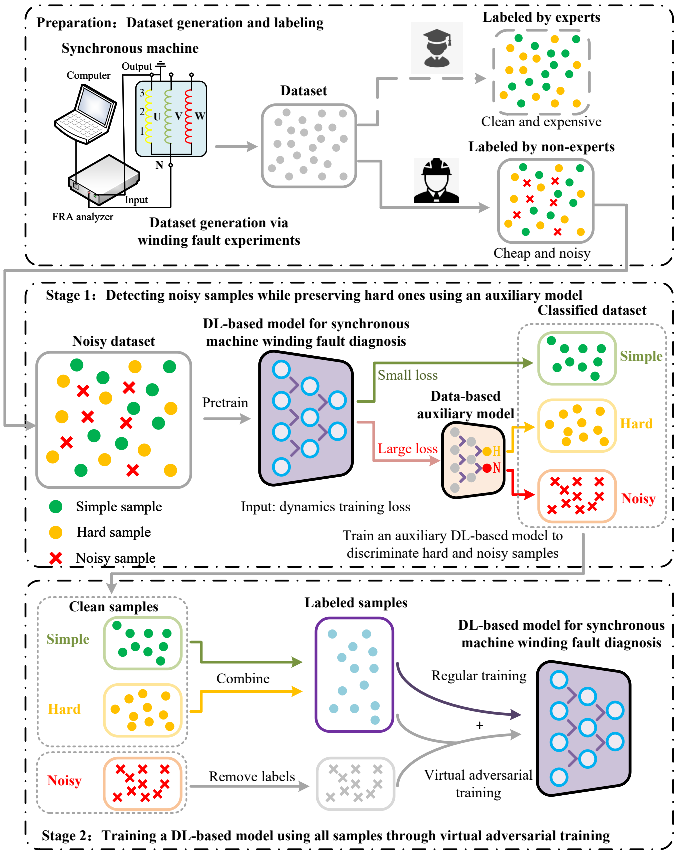

Fig. 5: Overview of the proposed two-stage WSL framework.

is applied to the hard/noisy dataset $\mathcal{D}_{h/n}$ using $L_{0,h/n}$ to split it into hard samples $\mathcal{D}_h$ and noisy samples $\mathcal{D}_n$. The process of Stage I is shown in Fig. 6, and the pseudocode is shown in Algorithm 1.

Stage II: Training a DL-based model using all samples

Following the sample discrimination in Stage I, Stage II aims to fully exploit all available samples, including clean samples with correct labels and those identified as noisy with labels removed, to train a robust winding SC fault diagnosis model. Rather than discarding noisy samples, this stage treats them as unlabeled samples and adopts VAT in a semi-supervised learning mode [33]. The core principle of VAT is to enhance model robustness by enforcing local smoothness in the output distribution. This is achieved by encouraging the model to produce consistent predictions when inputs are subjected to small perturbations. Formally, for each input sample x, VAT first identifies the virtual adversarial direction r_{vadv}, which is the perturbation within a small ϵ-ball (in the L_2 norm) that maximally changes the model's output distribution:

$$r_{\text{vadv}} = \arg \max_{r;|r|_2 \leq \epsilon} D\left[p(y|x,\theta), p(y|x+r,\theta)\right] \tag{5}$$

where $D[\cdot,\cdot]$ denotes the Kullback-Leibler (KL) divergence, which measures the difference between the original and perturbed output distributions. The model is then regularized by minimizing this maximum divergence:

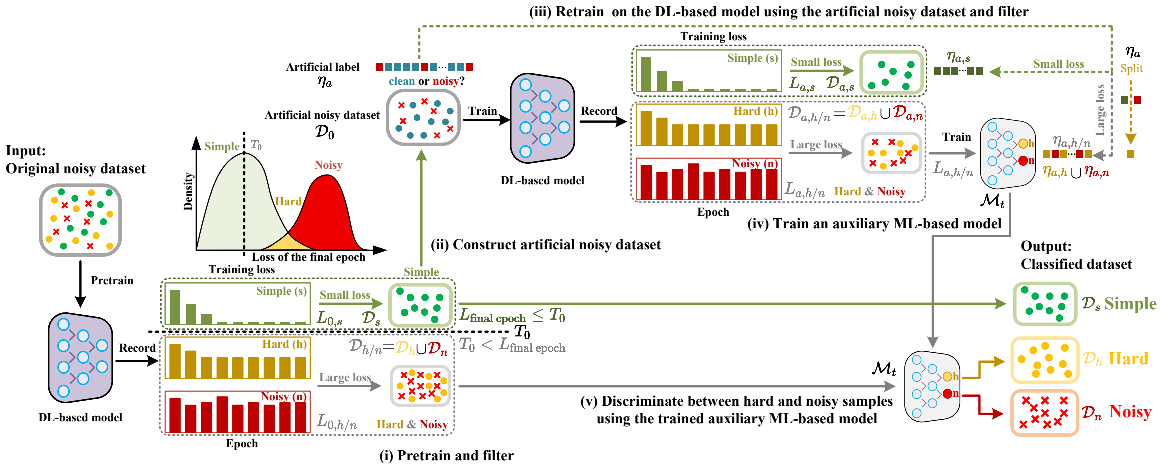

Fig. 6: Stage I of the proposed two-stage WSL framework.

Algorithm 1 Detecting noisy samples while preserving hard ones in Stage I

Require: Noisy dataset $\mathcal{D}$, DL-based winding SC fault diagnosis model $\mathcal{M}_0$, and small-loss criterion threshold T_0

Ensure: Confirmed simple samples $\mathcal{D}_s$, hard samples $\mathcal{D}_h$, and noisy samples $\mathcal{D}_n$.

1: **Step 1: Pretrain and Filter**
2: Train $\mathcal{M}_0$ on $\mathcal{D}$ while recording full dynamic training loss L_0
3: Split $\mathcal{D} \to \mathcal{D}_s$ (simple) and $\mathcal{D}_{h/n}$ (hard/noisy) via small-loss criterion threshold T_0
4: **Step 2: Construct Artificial Noisy Dataset**
5: Inject artificial label noise (ratio η_a) into $\mathcal{D}_s$ to form $\mathcal{D}_a$
6: **Step 3: Retrain on Artificial Noisy Dataset**
7: Train new model $\mathcal{M}_1$ on $\mathcal{D}_a$ while recording the per-sample dynamic training losses L_a
8: Extract $\mathcal{D}_{a,h/n}$, $L_{a,h/n}$, $\eta_{a,h/n}$ via small-loss criterion threshold T_0
9: **Step 4: Auxiliary Model Training**
10: Train auxiliary model $\mathcal{M}_t$ on $(L_{a,h/n}, \eta_{a,h/n})$
11: **Step 5: Sample Discrimination**
12: Apply $\mathcal{M}_t$ to $L_{0,h/n}$ to classify $\mathcal{D}_{h/n} \to \mathcal{D}_h$ (hard) and $\mathcal{D}_n$ (noisy)

$$L_{\text{VAT}}(x, \theta) = D\left[p(y|x, \theta), p(y|x + r_{\text{vadv}}, \theta)\right] \tag{6}$$

This regularization strategy effectively shifts the decision boundary away from high-density regions in the input space, yielding a smoother classification function with reduced sensitivity to minor input perturbations. As depicted in Fig. 7 and outlined in Algorithm 2, Stage II implements this approach by treating identified noisy samples as unlabeled and incorporating them into the VAT framework alongside clean samples. By utilizing both labeled and unlabeled samples through VAT, Stage II achieves dual objectives: maximizing the utilization of all available samples while enhancing the model's robustness against data and label noise.

Algorithm 2 Learning a DL-based model using all samples in Stage II

Require: Labeled samples $\mathcal{D}_l = \mathcal{D}_s \cup \mathcal{D}_h$, unlabeled samples $\mathcal{D}_{ul} = \mathcal{D}_n$, DL-based winding SC fault diagnosis model $\mathcal{M}^*$, training epoch N_{epoch}, number of labeled samples N_l, number of unlabeled samples N_{ul}, cross-entropy loss L_{clf}, hyperparameter α, and learning rate η

Ensure: Robust DL-based winding SC fault diagnosis model $\mathcal{M}^*$ with parameters θ

1: Initialize model parameters θ
2: Set VAT regularization coefficient α
3: **for** epoch = 1 to N_{epoch} **do**
4: Compute classification loss:
5: $L_{\text{clf}} = \frac{1}{N_l} \sum_{(x,y)\in\mathcal{D}_l} L_{\text{clf}}(y, p(y|x,\theta))$
6: Compute VAT regularization loss:
7: $L_{\text{VAT}} = \frac{1}{N_l+N_{ul}} \sum_{x\in\mathcal{D}_l\cup\mathcal{D}_{ul}} L_{\text{VAT}}(x,\theta)$
8: Update parameters: $\theta \leftarrow \theta - \eta\nabla_\theta(L_{\text{clf}} + \alpha L_{\text{VAT}})$
9: **end for**

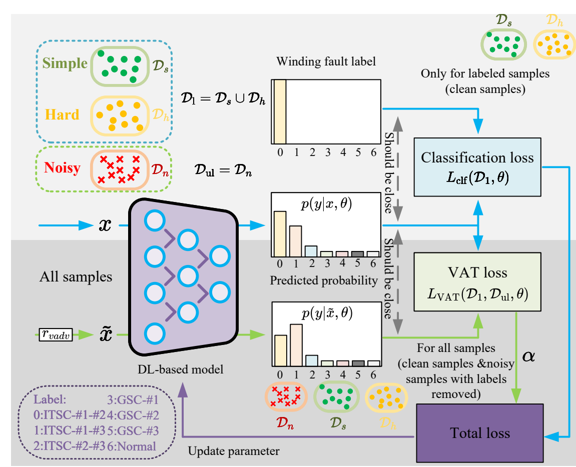

Fig. 7: Stage II of the proposed two-stage WSL framework.

III. ARTIFICIALLY SIMULATED WINDING SC FAULT EXPERIMENTS AND FRA DATASET

Previous studies have constructed an experimental platform to artificially simulate different synchronous machine winding

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TABLE I: Nameplate Specifications of the Synchronous Machine and FRA Analyzer [4]

Characteristics	Parameter value
Rated power	5 kVA
Rated voltage	380 V
Frequency	50 Hz
Pole pairs	1
Number of slots	36
Rated speed	1500 rpm
FRA analyzer	
Model	TDT6U
Manufacturer	Beijing Shengtai Real-Time
Output	25 Vpp (maximum, adjustable)
Output impedance	1 MΩ / 50 Ω (optional)
Frequency sweep range	1 Hz to 2 MHz
Frequency accuracy	99.995%
Dynamic range	-120 dB to 20 dB
Gain accuracy	± 0.5 dB
Test speed	no more than 1 minute

SC faults [4], [16]. The experimental platform comprises an FRA analyzer and a 5 kVA synchronous machine. The nameplate specifications for the associated equipment are detailed in Table I.

TABLE II: FRA Dataset of Synchronous Machine Winding [4]

Fault Type	Fault Severity (Resistance)	Total Samples
GSC-#1	0, 0.1, 10, 20, 40 Ω[a]	974
GSC-#2		1002
GSC-#3		1001
ITSC-#1-#2		1002
ITSC-#1-#3		999
ITSC-#2-#3		1002
Normal	N/A	200

[a] Each fault class aggregates multiple resistance levels, while the normal class has no severity subdivision. Approximately 200 samples are collected for each resistance value.

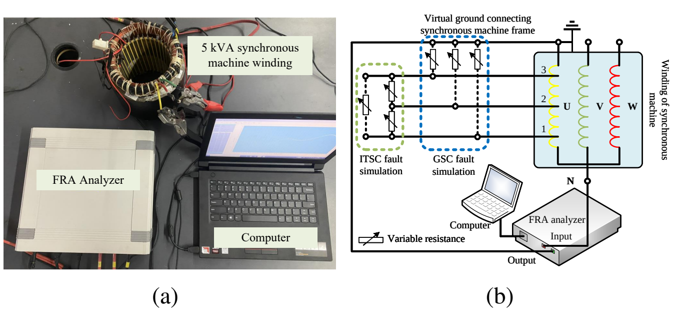

Fig. 8: Measurement experimental diagram [4]. (a) Actual wiring diagram. (b) Measurement wiring diagram.

Referring to prior studies [4], [16], [24], [34], [35], multiple winding SC faults are simulated. Specifically, GSC faults are simulated by shorting winding slots 1, 2, and 3 of the U-phase to ground, referred to as GSC-#1, #2, and #3, respectively. ITSC faults are simulated by interconnecting these slots in pairs, referred to as ITSC-#1-#3, #1-#2, and #2-#3. Fault severity is simulated using series resistors of 0, 0.1, 10, 20, and 40 Ω, where lower resistance values correspond to more severe winding SC faults [34], [35]. The experimental wiring configuration, shown in Fig. 8, employs an end-to-end open-circuit connection. Input and output signals are recorded and processed via Fourier transformation to derive FRA gain and phase data, as defined by Eqs. (1)-(2). Repeated measurements yield an FRA dataset comprising seven fault categories, with 6,180 samples, as summarized in Table II. Representative FRA gain and phase data for typical winding SC faults are shown in Fig. 9. For more experimental details and related settings, readers can refer to Refs. [4], [16], [24], [32]. It should be noted that this study uses resistors ranging from 0.1 to 40 Ω to simulate different severities of synchronous machine winding SC faults, solely to validate the proposed method. In reality, resistance values must be determined by the specific machine characteristics and actual fault conditions [36], [37].

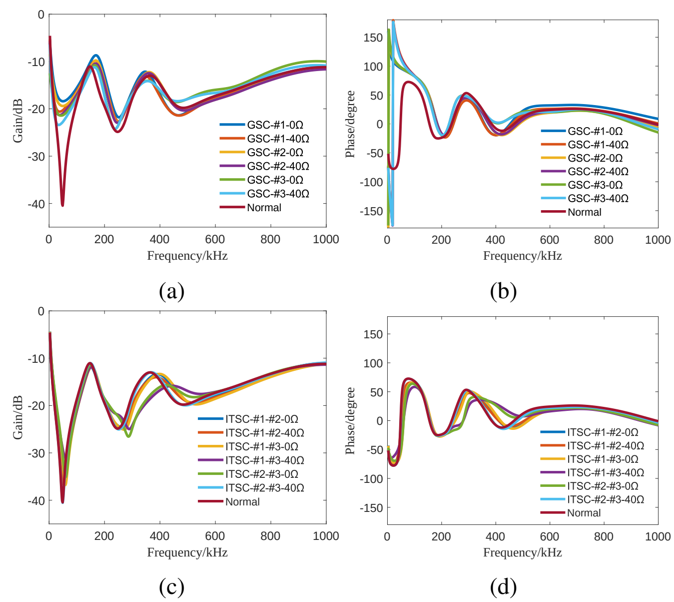

Fig. 9: Representative FRA data of typical winding SC faults [4]. (a) FRA gain of GSC. (b) FRA phase of GSC. (c) FRA gain of ITSC. (d) FRA phase of ITSC.

IV. EXPERIMENTS

Three types of experiments are designed to evaluate the proposed method: the first validates data noise mitigation via binary morphology; the second assesses label noise mitigation through the two-stage WSL; and the third examines the overall performance under combined data and label noise.

It should be noted that all performance metrics reported in this section are evaluated on a test set comprising 30% of the total samples. The experiments are conducted using the computational environment specified in Table III. The architecture of the DL-based winding SC fault diagnosis model is illustrated in Fig. 10, whose core components have been validated as effective in Ref. [4]. Additionally, the complete source code and hyperparameter settings are made publicly available in the associated GitHub repository [1].

[1] https://github.com/cy1034429432/Learning_from_data_and_label_noise

TABLE III: Computer Hardware and Software Configuration

Device or software	Nameplate value or configuration
CPU	Intel Ultra 7 265KF
GPU	RTX 5060 Ti
RAM	32 GB
Software	Python
Packages	PyTorch, scikit-learn, NumPy

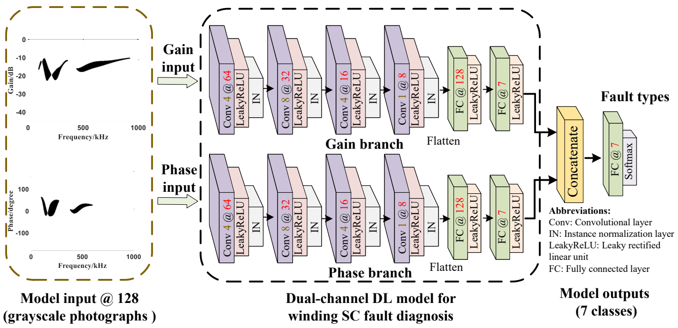

Fig. 10: Architecture of the synchronous machine DL-based winding SC fault diagnosis model [4]. The numbers following “Conv” and “@” denote the number of output channels and the spatial dimensions of the feature maps, respectively, with height equal to width.

A. Experimental Validation for Data Noise

The FRA measurement process may introduce data noise, such as cabling-induced noise, sensor errors, and electromagnetic interference. As the experimental data are acquired under controlled laboratory conditions over a continuous period, the inherent noise in the raw measurements is considered negligible. To emulate realistic data noise, artificial Gaussian noise is introduced into the measured FRA gain and phase data [38], [39]. The signal-to-noise ratio (SNR) is adopted to quantify the noise intensity, defined as the ratio between the signal power and noise power, commonly expressed in decibels (dB):

$$\mathrm{SNR} = 10\log_{10}\left(\frac{P_s}{P_n}\right) \tag{7}$$

where P_s and P_n represent the power of the signal and noise, respectively. The noise injection procedure is implemented as follows: first, the signal power is calculated as $P_s = \frac{1}{N}\sum_{i=1}^{N}|x_i|^2$; then, the noise power is determined by $P_n = Ps/10^{SNR/10}$; finally, Gaussian noise $n \sim \mathcal{N}(0,\sqrt{P_n})$ is generated and added to the original data, yielding $y = x + n$. A lower SNR value corresponds to a higher degree of data noise. In this study, SNR is used only to control the injected additive data noise. For example, SNR = 10 dB corresponds to $P_n = 0.1P_s$, i.e., the noise root-mean-square amplitude is approximately 31.6% of the signal root-mean-square amplitude. Since the FRA data are obtained from an offline sweep frequency measurement rather than from continuously operating time-domain signals, this level already represents a visibly disturbed FRA curve in the experimental setting. This interpretation is also consistent with the instrumentation specifications in Table I, where the gain accuracy is within ±0.5 dB, and the frequency accuracy reaches 99.995%.

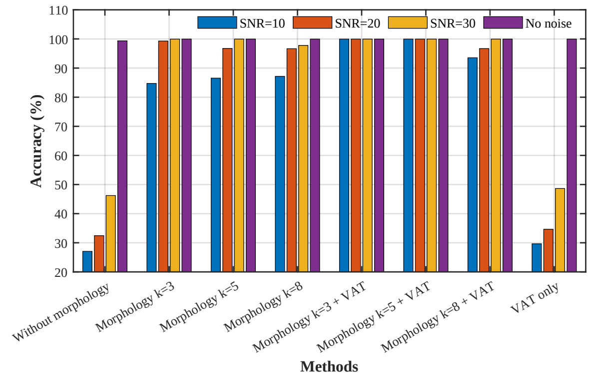

Fig. 11: Comparison of test set accuracies under varying data noise levels. The experiments are performed by training the model on the original noise-free dataset and evaluating its performance on test sets with different degrees of data noise.

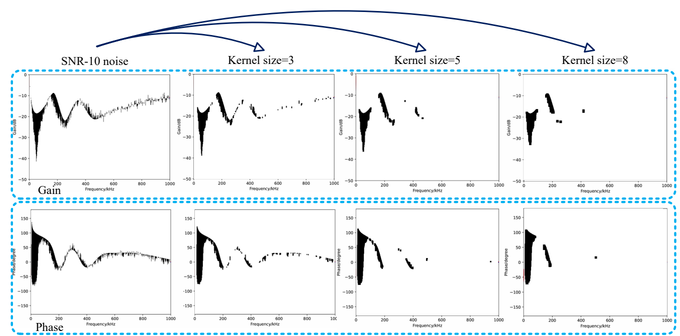

Fig. 12: Representative visualization results after binary morphology under different kernel sizes.

This study employs the original FRA dataset, regarded as a noise-free dataset, to train the DL-based winding SC fault diagnosis model under different binary morphology settings (i.e., kernel sizes), while evaluating its performance on test sets within varying degrees of data noise. The quantitative results and representative visualizations, shown in Figs. 11-12, respectively, lead to the following observations: (1) Models trained on a noise-free dataset exhibit pronounced performance degradation as the noise degree increases, indicating that intrinsic robustness is insufficient to cope with data noise. (2) Binary morphology markedly improves noise resistance, suggesting that laboratory data can be leveraged to train models with strong out-of-domain generalization when suitable preprocessing is applied. (3) The kernel size for binary morphology must be tuned to the image scale and noise characteristics. An overly large kernel may erase subtle variations in some frequency bands, whereas an overly small kernel fails to adequately suppress noise. (4) Binary morphology applied to noisy data leaves residual speckle. Incorporating VAT substantially enhances robustness to these speckles and yields significant accuracy improvement, as VAT introduces similar speckle perturbations (i.e., r_{vadv}) during training. By contrast, applying VAT directly to the original two FRA images (FRA gain-curve and FRA phase-curve data) without binary morphology yields limited benefits. Besides, it should be noted that lower-SNR conditions may require an additional front-end denoising stage before binary morphology. Therefore, the present results should be interpreted as validation under controlled offline FRA noise conditions, while extremely low-SNR field measurements remain an important topic for future work.

In addition, this study tests two key hyperparameters in VAT: ϵ, which determines the magnitude of the adversarial perturbation r_{vadv}, and α, which specifies the weight of the regularization term, while fixing the kernel size at 5 in binary morphology. The average accuracies across different data noise degrees are reported in Table IV. The results indicate that optimal performance is obtained with moderate perturbation strength ($\epsilon \approx 0.1 - 0.2$) and moderate regularization weight ($\alpha \approx 0.2 - 0.3$). The accuracy declines when ϵ is either too small, leading to weak perturbations, or too large, causing training instability. Similarly, performance deteriorates when α is too low because regularization is insufficient, or when α is too large because the classification objective is over-constrained, which is consistent with Ref. [15], [33].

TABLE IV: Ablation Study of VAT Hyperparameters (ϵ and α)

Accuracy	$\epsilon = 0.01$	$\epsilon = 0.1$	$\epsilon = 0.2$	$\epsilon = 0.5$
$\alpha = 0.1$	92.15%	91.78%	93.34%	88.67%
$\alpha = 0.2$	87.05%	**99.95%**	98.17%	90.34%
$\alpha = 0.3$	90.40%	99.62%	**99.95%**	93.56%
$\alpha = 0.4$	85.00%	89.64%	93.05%	95.80%

B. Experimental Validation for Label Noise

Hiring domain-specific experts to label samples is costly. Consequently, the labeling of synchronous machine fault samples is often performed by non-expert engineers, inevitably introducing label noise. Such noise is typically categorized as either feature-dependent or feature-independent. Feature-dependent noise originates from the complexity and inherent ambiguity of synchronous machine fault characteristics. Its distribution is closely associated with the underlying data features, and its probability can be modeled as follows:

$$P(\tilde{y} = j \mid y = i, x) = \rho_{i,j}(x) \tag{8}$$

where $\tilde{y}$ denotes the noisy label, y represents the true label, and x corresponds to the input data. The noise transition probability $\rho_{i,j}(x)$ depends on both the true label i and the input data x. In this study, feature-dependent label noise primarily occurs when non-expert engineers mislabel fault locations within the same fault category, excluding the normal class. Typical examples include mislabeling GSC-#1 as GSC-#2 or ITSC-#1-#2 as ITSC-#1-#3, as shown in Fig. 13. In these examples, the corresponding FRA data exhibit high similarity, making them difficult to distinguish. Except for experimental designers or relevant experts, most engineers are unable to discern such differences. Let ρ_{dep} represent the proportion of feature-dependent label noise present in the training samples.

Feature-independent noise results from factors unrelated to data features, primarily stemming from human annotation

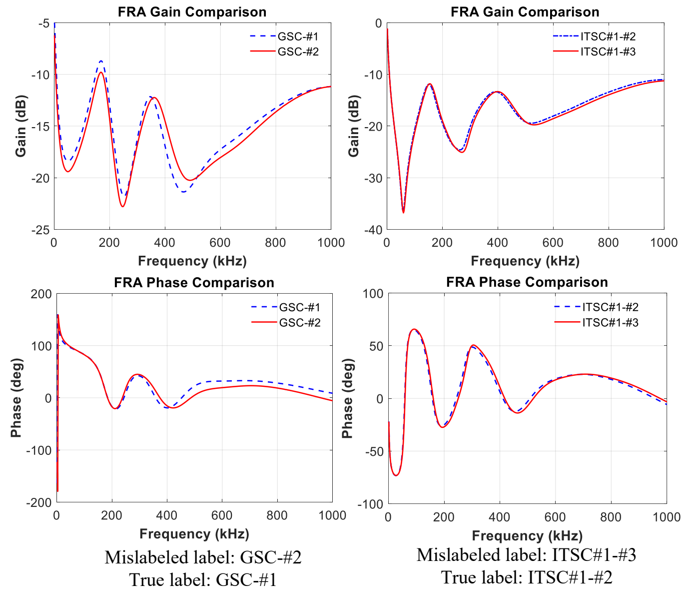

Fig. 13: Representative examples for feature-dependent noise.

errors. In synchronous machine fault diagnosis, this type of noise appears as randomly distributed label errors caused by annotator mistakes, with occurrence probabilities independent of the specific patterns or distributions of fault characteristics. Mathematically, feature-independent label noise can be modeled as:

$$P(\tilde{y} = j \mid y = i) = \rho_{i,j} \tag{9}$$

where the noise transition probability $\rho_{i,j}$ is only related to the true label i and the incorrect label j, and is independent of the input data x. In this study, feature-independent label noise is modeled by randomly assigning, with uniform probability, one of the incorrect labels from the remaining classes. Let ρ_{ind} represent the proportion of feature-independent label noise present in the training samples.

Fig. 14 presents the test set accuracy under different label noise degrees. It can be seen that: (1) Without appropriate processing, model performance degrades progressively as the proportion of samples with incorrect labels increases. (2) With the two-stage WSL, model performance remains robust to increasing label noise degree. Moreover, VAT further enhances stability and accuracy. (3) Compared with the commonly adopted Co-teaching method [40] and small-loss criterion [27], the proposed method demonstrates consistent superiority across all label noise degrees.

It should be noted that, in Stage I, the proposed method employs the small-loss criterion twice with distinct thresholds: 0.2 for the first and 0.5 for the second. This design reflects the fixed training epochs: during the first session, the larger number of samples enables sufficient model updates, whereas in the second session, the limited number of samples slows loss reduction, necessitating a higher threshold. With this setting, experiments using SVM to classify noisy and hard samples in an artificial dataset consistently achieve nearly 100% accuracy. To elucidate this superior performance, Fig. 15 visualizes the average training losses of simple, hard, and noisy

TABLE V: Test Set Accuracies under Combined Training Data Noise and Training Label Noise on the 5 kVA Synchronous Machine FRA Dataset

$\rho_{ind} + \rho_{dep}$ =[a]	10%	20%	30%	40%
Training dataset with SNR = 10 dB	99.24%(±0.51%)	98.13%(±0.60%)	97.13%(±1.08%)	95.13%(±1.98%)
Training dataset with SNR = 20 dB	99.31%(±0.45%)	98.14%(±0.56%)	97.34%(±0.82%)	95.38%(±1.79%)
Training dataset with SNR = 30 dB	99.44%(±0.42%)	98.19%(±0.54%)	97.55%(±0.83%)	95.72%(±1.80%)

[a] ρ_{ind} and ρ_{dep} denote the proportions of feature-independent and feature-dependent label noise, respectively. Their sum represents the total proportion of training samples with incorrect labels. For a given total label noise level, the two components can be assigned with different proportions.

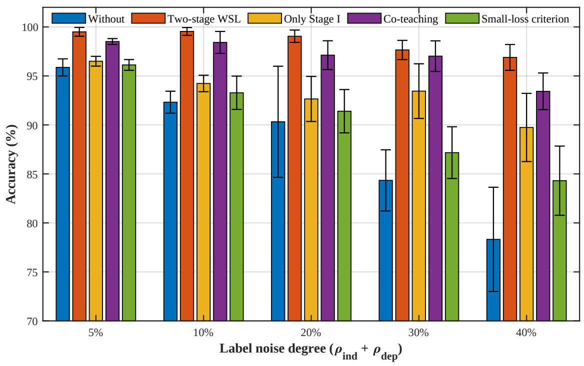

Fig. 14: Comparison of test set accuracies with standard deviation under different label noise degrees ($\rho_{ind} + \rho_{dep}$). Without data noise, the model is trained and evaluated on the training and test sets, respectively.

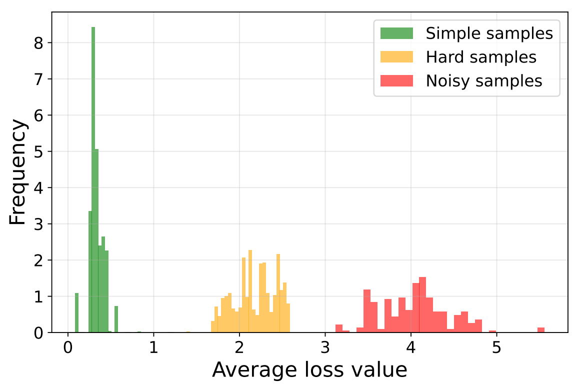

Fig. 15: Representative histogram of average training losses for simple, hard, and noisy samples identified by Algorithm 1, where simple samples are selected using the initial small-loss threshold and hard/noisy samples are separated by the auxiliary SVM.

samples, revealing clearly differentiated loss characteristics that mechanistically explain the effectiveness of the proposed method in distinguishing among them.

C. Experimental Validation for Combined Data and Label Noise

The preceding sections evaluate data noise and label noise separately. However, these two types of noise can be coupled. For instance, data noise may cause FRA measurements to resemble those of a different winding state, leading annotators to assign an incorrect label.

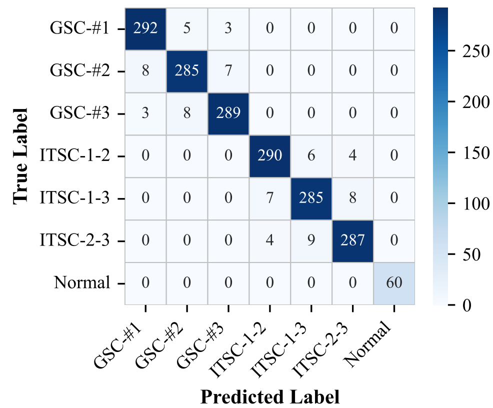

Fig. 16: Representative confusion matrix for the 5 kVA synchronous machine dataset with 40% label noise and 10 dB SNR data noise.

Therefore, this section evaluates the robustness of the proposed method when both data and label noise coexist in the training dataset, with corresponding results provided in Table V. In Table V, the row label "Training dataset with SNR-X" indicates that additive Gaussian data noise is injected into the training samples according to Eq. (7), where lower SNR indicates stronger data noise. The column label $\rho_{ind} + \rho_{dep}$ denotes the total proportion of training samples with incorrect labels, including both feature-independent and feature-dependent label noise. Experimental results demonstrate the method's effectiveness in mitigating both data and label noise, which can be attributed to three key factors: (1) Binary morphology enhances the intrinsic fault detection capability of FRA by highlighting salient fault characteristics and suppressing less salient or noise-dominated ones, thereby improving feature discriminability and noise resistance. (2) Two-stage WSL identifies noisy samples to prevent erroneous model mappings; subsequently, VAT is applied to exploit all samples, including noisy ones, thereby improving data utilization and model generalization. (3) Fixed kernel sizes in binary morphology may yield residual speckles owing to diverse fault characteristics or data noise degrees. VAT employs adversarial perturbations to desensitize the model to local speckles in the input, enhancing its robustness and effectively complementing binary morphology.

To complement the overall accuracy results, one representative confusion matrix is provided to illustrate class-wise performance under 40% label noise with a training dataset at 10 dB SNR. As shown in Fig. 16, for the 5 kVA synchronous machine dataset with seven categories, most classes are correctly identified, with only minor misclassifications among fault categories with similar FRA characteristics.

D. Performance Evaluation of the Proposed Method on a 7.5 kVA Synchronous Machine

TABLE VI: Specifications of the 7.5 kVA Synchronous Machine and Corresponding FRA Dataset [20]

Machine Specifications		
Characteristics	Parameter value	
Rated power	7.5 kVA	
Rated voltage	380 V	
Rated excitation voltage	32 V	
Rated excitation current	1.2 A	
Frequency	50 Hz	
Pole pairs	2	
Rated speed	1500 rpm	
Corresponding FRA dataset		
Fault type	Fault severity	Total samples
ITSC-Head position	0–10 Ω[a]	500
ITSC-Middle position		500
ITSC-End position		500
Normal	N/A	50

[a] Each fault class aggregates multiple resistance levels, while the normal class has no severity subdivision.

To evaluate the generalization and transferability of the proposed method, similar experiments are conducted on another 7.5 kVA synchronous machine, with detailed parameters provided in Table VI. Unlike the 5 kVA machine discussed in Section III, the measurements on the 7.5 kVA machine include the rotor. All FRA measurements in this study are performed offline with the machine shut down. To ensure measurement consistency, the rotor position remains fixed during each measurement, since variations in rotor position may introduce discrepancies in the FRA data [16], [19], [20]. In addition, only ITSC fault experiments and measurements under normal winding conditions are conducted, following procedures similar to those in Section III. The ITSC faults involved three locations, with detailed experimental settings provided in Ref. [20]. The experimental site is shown in Fig. 17, and the corresponding FRA dataset is summarized in Table VI.

Using the 7.5 kVA synchronous machine FRA dataset and the previously selected parameter settings without further tuning, two representative experiments are conducted: (1) comparison of different methods under varying label noise degrees (Fig. 18), with loss function values from one typical training process shown in Fig. 19; and (2) evaluation of the proposed method under different degrees of combined data and label noise (Table VII). Results from the 7.5 kVA synchronous machine FRA dataset, consistent with previous findings, confirm that the proposed method effectively identifies mislabeled samples and mitigates the impact of both data and label noise.

For the 7.5 kVA dataset with 40% label noise and 10 dB SNR data noise, the representative confusion matrix in Fig. 20 demonstrates that ITSC faults at different positions (head, middle, end) are generally well separated, although some confusion remains between ITSC-Head and ITSC-End. Importantly, the normal class in Fig. 16 and Fig. 20 is consistently recognized with high reliability, despite the class imbalance (i.e., fewer normal samples compared to other classes) shown in Tables II and VI. These results provide a clearer view of the diagnostic capability across confusing classes, thereby complementing the accuracy metrics and enhancing the transparency of the evaluation.

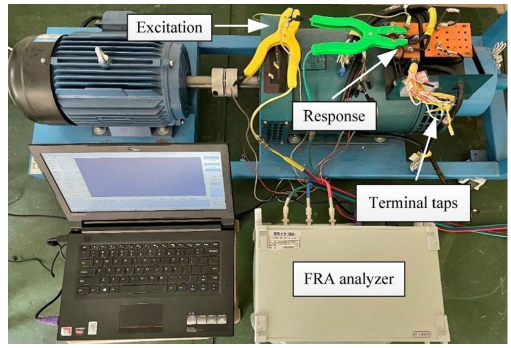

Fig. 17: Experimental wiring diagram of the 7.5 kVA synchronous machine. The location and severity of ITSC faults are determined by selecting different terminal taps and resistors with varying resistance values.

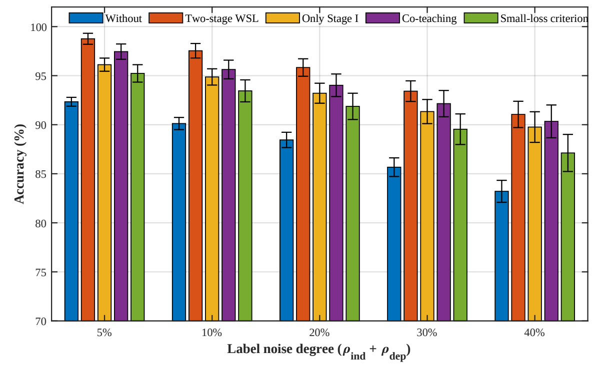

Fig. 18: Comparison of test set accuracies with standard deviation under different label noise degrees ($\rho_{ind} + \rho_{dep}$) on the 7.5 kVA synchronous machine FRA dataset.

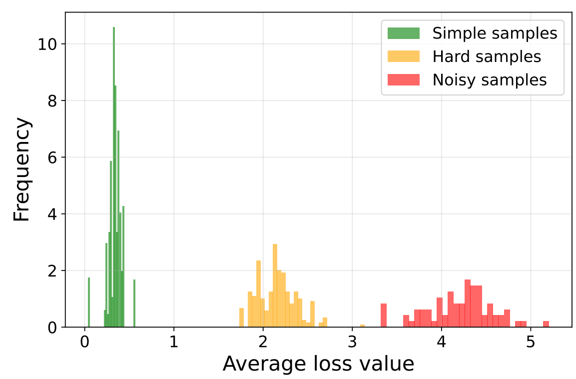

Fig. 19: Representative histogram of average training losses of simple, hard, and noisy samples on the 7.5 kVA synchronous machine FRA dataset.

TABLE VII: Test Set Accuracies under Combined Training Data Noise and Training Label Noise on the 7.5 kVA Synchronous Machine FRA Dataset

$\rho_{ind} + \rho_{dep} =$	10%	20%	30%	40%
Training dataset with SNR = 10 dB	96.84%(±0.72%)	95.12%(±0.95%)	93.42%(±1.42%)	91.12%(±2.05%)
Training dataset with SNR = 20 dB	97.05%(±0.65%)	95.34%(±0.88%)	93.63%(±1.48%)	91.38%(±1.92%)
Training dataset with SNR = 30 dB	97.18%(±0.58%)	95.52%(±0.92%)	93.85%(±1.35%)	91.72%(±1.88%)

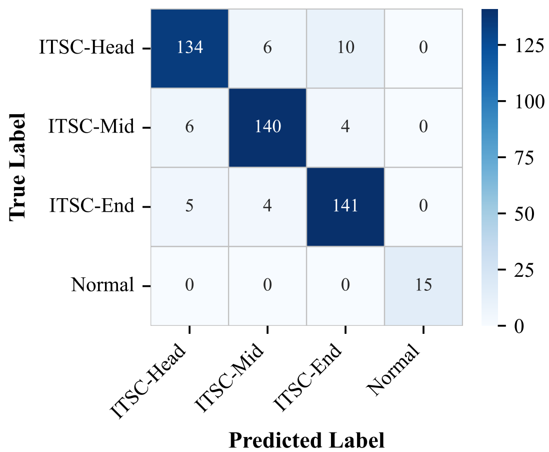

Fig. 20: Representative confusion matrix for the 7.5 kVA synchronous machine dataset with 40% label noise and 10 dB SNR data noise.

V. Discussion and Limitations

While numerous studies have proposed data-driven SC fault diagnosis methods for synchronous machine windings, most studies prioritize diagnostic accuracy without accounting for the impact of data and label noise. In practical fault diagnosis, data-driven models face distinct challenges: data noise arising from measurement conditions and label noise caused by similar fault characteristics or human factors. This study further highlights the coupling between data and label noise. Specifically, data noise often directly induces label noise, whereas suppressing data noise reduces the possibility of label noise. Hence, the first step in dealing with label noise is to mitigate data noise, ensuring the training dataset contains two distinguishable types: feature-dependent and feature-independent label noise. In practice, the training datasets are typically supplied by product manufacturers or power grid companies rather than collected in laboratories or annotated by experts. Consequently, both data and label noise are inevitably present. The proposed method thus holds promise for application to real FRA datasets.

Current routine FRA-based inspections of synchronous machine windings depend heavily on subjective expert judgment. While the proposed method makes this process more objective and effectively mitigates the impact of data and label noise, it remains subject to the following limitations:

1) The validation in this study is limited to offline FRA measurements under controlled laboratory conditions and therefore does not constitute validation of online diagnosis under time-varying-speed operation, where nonstationary electromagnetic and mechanical interference may contaminate the measured signals. Since collecting industrial datasets with reliable winding fault labels and controllable noise sources is difficult and potentially unsafe, owing to the cost and risk of inducing faults in operating machines, this study uses two laboratory machines with controlled data and label noise injection to verify the core learning mechanism. The results show that the proposed binary morphology and two-stage WSL framework improve robustness to controlled data and label noise in offline FRA-based synchronous machine winding diagnosis. However, validation on field datasets and extension to online diagnosis with adaptive front-end denoising remain necessary before deployment in time-varying industrial environments [39], [41], [42].
2) The results in Figs. 11-12 and Table IV indicate that both the binary morphology module and VAT require appropriate parameter selection. For binary morphology, the kernel size is closely related to the data noise level and measurement conditions: an excessively large kernel may smooth subtle frequency-band variations and cause information loss, whereas an overly small kernel may provide insufficient noise suppression. This limitation is consistent with Ref. [13]. For VAT, although extreme settings degrade performance, Table IV identifies a stable practical range of $\epsilon = 0.1 - 0.2$ and $\alpha = 0.2 - 0.3$. The final parameters are selected within this moderate range and transferred unchanged to the independent 7.5 kVA synchronous machine dataset in Section IV-D. The consistent results in Figs. 18-20 and Table VII suggest that the proposed method does not rely on case-specific exhaustive tuning, although a small validation subset is still recommended when deploying it to a new machine. Future work will explore adaptive parameter selection mechanisms that adjust the morphology kernel and VAT settings according to noise intensity and measurement conditions, thereby improving robustness and reducing manual intervention.
3) The current validation of noise-induced mislabeling still relies on simulated perturbations and synthetic label corruption because real field data with known noise sources and verified winding fault labels are unavailable. To partially address this limitation, cable-induced noise is simulated using the circuit model in Fig. 1. As shown in Fig. 21, normal-state samples affected by such noise may resemble ITSC-#2-#3 and therefore be mislabeled. The similarity between these simulated perturbations and the scenarios in Fig. 13 suggests that the proposed method is applicable to this type of noise-induced mislabeling, although further validation using field data remains nec-

essary.

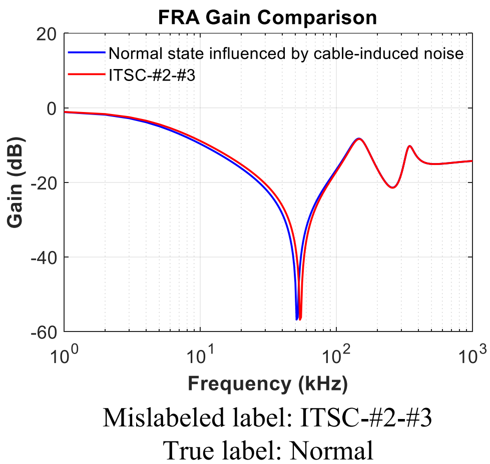

Fig. 21: An example of mislabeling induced by data noise. Stray inductance and capacitance are introduced in the winding equivalent circuit to simulate cable-induced noise.

VI. Conclusion

To mitigate data and label noise in synchronous machine winding SC fault diagnosis, this study proposes a robust method for SC fault diagnosis based on binary morphology and two-stage WSL. Based on the experimental results, the following conclusions are drawn:

1) DL-based models exhibit limited inherent robustness to data noise, while their performance degrades significantly with increasing label noise. Given the coupling between these noises, mitigating data noise is a prerequisite for effectively addressing label noise.
2) The integration of binary morphology with FRA data effectively mitigates data noise: erosion suppresses noise across all frequency bands, while dilation restores informative bands that should be preserved. Furthermore, VAT mitigates the residual speckle caused by binary morphology using a fixed kernel size, thereby enhancing the modelâs robustness to data noise.
3) For label noise, an auxiliary model guided by the small-loss criterion can distinguish hard samples from noisy ones, enabling the use of challenging yet informative samples that are often discarded by prior methods. In addition, VAT, as a semi-supervised learning method, incorporates all samples into training by treating noisy ones as unlabeled, thereby substantially improving data utilization and model robustness.

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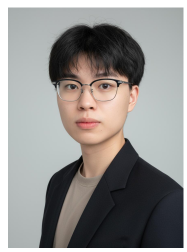

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